Retained adjoint-fermion binding with explicit $\hbar$: recoil, quantum corrections and the whole-state current

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Abstract

We study two fermionic endpoints bound to a retained adjoint fermion by regulated contact interactions, with no direct quark–antiquark force. In a fixed-number, global-color model, dimer binding suffices to prove a trimer below breakup. With physical units and explicit $\hbar$, we distinguish recoil perturbation theory, the semiclassical expansion and contact iterations. The leading recoil energy correction is second order and attractive. For a Gaussian contact in the deep-well regime, we derive the energy and whole-state charge radius through order $\hbar^2$, with exact mass-ratio dependence at each displayed order. The leading anharmonic radius correction is positive. Contact binding does not by itself make this current pointlike. A formal π/K scalar pair supplies the charged Dirac muon multiplet count, while exact physical $B - L$ imposes a further conditional charge constraint. We compare a distinct 1983 composite-fermion mass-protection mechanism and retain endpoint and Goldstino proposals as construction hypotheses. A common string realization remains a separate task.

Research scope and provenance. The broader target remains bosonic strings with $q/\bar{q}$ or q/q endpoints for mesons or diquarks, and fermionic strings with $q/\bar{q}$ or $\bar{q}/\bar{q}$ endpoints for leptons or quarks. The main phenomenological question is approximately pointlike scattering of the whole fermionic state. The user now selects the retained adjoint contact model as the most interesting concrete object and requests explicit $\hbar$ in all calculations.

This focused revision develops that selection. It does not identify a contact trimer with a string by definition. Version 0.2, including the warped-current comparison and the scalar-completion study, remains preserved at repository commit `6b73d29`; version 0.1 is preserved at `333e537`. The uploaded originals remain unchanged. Earlier natural-unit calculations are historical comparisons, not the unit convention of this paper. The new radius calculation uses standard perturbation theory; no claim of literature priority follows from its application here. The later hypothesis section does not change the fermionic-endpoint Hamiltonian used for this calculation.

1 A retained adjoint binds both endpoints

We take distinguishable nonrelativistic endpoint species $q, \bar{q}$ with physical masses m, m, and a retained adjoint fermion λ with physical mass M. Constituent numbers are conserved and color is global $SU(3)$. These assumptions define the model, not a relativistic QCD completion. The carrier is present in the bound state; it is not a virtual fermion exchanged between unchanged endpoints.

Throughout, $\hbar$ is explicit, c_{light} denotes the speed of light, positions have units of length and momenta have units of momentum:

$$\mathbf{p} = -i\hbar\nabla, \qquad i\hbar\partial_t\Psi = H\Psi, \qquad [G] = \text{energy}\,\text{length}^3. \tag{1}$$

The color matrices are dimensionless and obey $\mathrm{tr}(T^a T^b) = \delta^{ab}/2$. The unique singlet in $\mathbf{3} \otimes \bar{\mathbf{3}} \otimes \mathbf{8}$ has

$$T_q \cdot T_{\bar{q}} = \frac{1}{6}, \qquad T_q \cdot T_\lambda = T_{\bar{q}} \cdot T_\lambda = -\frac{3}{2}. \tag{2}$$

There is no direct $q\bar{q}$ potential. With $\mathbf{x} = \mathbf{r}_q - \mathbf{r}_\lambda$ and $\mathbf{y} = \mathbf{r}_{\bar{q}} - \mathbf{r}_\lambda$, set

$$f_R(r) = \frac{e^{-r^2/R^2}}{\pi^{3/2} R^3}, \qquad V = -\frac{3G}{2}[f_R(x) + f_R(y)], \qquad G > 0, \tag{3}$$

$$\mu = \frac{mM}{m+M}, \qquad H_{\mathrm{rel}} = \frac{p_x^2 + p_y^2}{2\mu} + \frac{\mathbf{p}_x \cdot \mathbf{p}_y}{M} + V. \tag{4}$$

Equivalently the kinetic operator is $-\hbar^2(\nabla_x^2 + \nabla_y^2)/(2\mu) - \hbar^2 \nabla_x \cdot \nabla_y / M$. The two contacts are number-conserving four-fermion density interactions. No propagating bosonic binding field is included in this Hamiltonian. Their ultraviolet origin remains unspecified. $R > 0$ is the specified finite regulator length; we do not silently take an unrenormalized three-dimensional delta limit.

The spin-independent model allows the two spin-1/2 endpoints to form a spin singlet, with total spin 1/2 supplied by λ. Each constituent has $s^2 = 3\hbar^2/4$; the endpoints here are not scalar particles. Spin-dependent interactions, allowed annihilation channels and gauged color would change the problem and must be supplied explicitly.

1.1 Binding and the correct breakup threshold

For $h_\mu = p^2/(2\mu) - 3G f_R/2$, a normalized Gaussian orbital of width a gives

$$E_2^{\mathrm{trial}}(a) = \frac{3\hbar^2}{4\mu a^2} - \frac{3G}{2\pi^{3/2}(R^2 + a^2)^{3/2}}. \tag{5}$$

At $a^2 = 2R^2$ this proves the sufficient condition

$$\boxed{G > \frac{3\sqrt{3}\,\pi^{3/2}}{4} \frac{\hbar^2 R}{\mu} \quad \Longrightarrow \quad \epsilon_2 < 0.} \tag{6}$$

Here ϵ_2 is the exact dimer ground energy, not its variational estimate. If u is its real normalized orbital, the product $u(x)u(y)$ has $\langle \mathbf{p}_x \cdot \mathbf{p}_y \rangle = 0$ and energy $2\epsilon_2$. Both dimer-plus-spectator thresholds are ϵ_2, so

$$\boxed{E_3 \le 2\epsilon_2 < \epsilon_2 < 0.} \tag{7}$$

The bounded short-range potentials and variational principle establish a trimer below breakup. Removing λ removes both interactions. This is a binding result with no pre-existing $q\bar{q}$ bound state.

Nonrelativistic control requires characteristic momenta much smaller than mc_{light} and Mc_{light}, and binding energies small compared with the corresponding rest energies. A useful range choice is $R \gg \hbar/(mc_{\mathrm{light}}), \hbar/(Mc_{\mathrm{light}})$. The model does not yet generate a light relativistic lepton, a confining flux tube, or protection against annihilation in a larger theory.

2 Three different expansions

Define a depth, a dimensionless contact strength, and a recoil parameter:

$$V_0 = \frac{3G}{2\pi^{3/2} R^3}, \qquad \gamma = \frac{2\mu V_0 R^2}{\hbar^2} = \frac{3\mu G}{\pi^{3/2}\hbar^2 R}, \qquad \eta = \frac{\mu}{M} = \frac{m}{m+M}. \tag{8}$$

The energy unit $E_R = \hbar^2/(2\mu R^2)$ and coordinates $\mathbf{X} = \mathbf{x}/R$, $\mathbf{Y} = \mathbf{y}/R$ give

$$\frac{H_{\rm rel}}{E_R} = -\nabla_X^2 - \nabla_Y^2 - 2\eta\nabla_X \cdot \nabla_Y - \gamma(e^{-X^2} + e^{-Y^2}). \tag{9}$$

This rescaling displays $\hbar$ in E_R and γ; it does not set it to one. Equation (6) becomes $\gamma > 9\sqrt{3}/4$.

Three expansions must be kept distinct: a recoil series in η about two bound dimers; a deep-well semiclassical series in $\varepsilon = \gamma^{-1/2} \propto \hbar$ at fixed physical potential and masses; and a Born series in the contact interaction about free propagation. They have different domains. In particular, the existence of a shallow bound-state pole generally requires resumming the Born series.

3 Recoil perturbation theory about the bound product

At fixed μ, G, R write

$$H(\eta) = H_0 + \eta W, \qquad H_0 = h_\mu(x) + h_\mu(y), \qquad W = \frac{\mathbf{p}_x \cdot \mathbf{p}_y}{\mu}. \tag{10}$$

The unperturbed orbital ground state is $|0\rangle = |u, u\rangle$, with $E_0 = 2\epsilon_2$. Spin degeneracy is inert in this spin-independent Hamiltonian. Reversing $\mathbf{x}$ alone maps $H(\eta)$ unitarily to $H(-\eta)$. Thus the isolated orbital ground-energy branch is even near $\eta = 0$:

$$E_3(\eta) = 2\epsilon_2 - \eta^2\mathcal{C}_2 + O(\eta^4), \qquad \mathcal{C}_2 = \frac{1}{\mu^2}\sum_{a,b}\frac{|\langle a, b|\mathbf{p}_x \cdot \mathbf{p}_y|u, u\rangle|^2}{\Delta_a + \Delta_b} > 0. \tag{11}$$

Here $\Delta_a = \epsilon_a - \epsilon_2$; the spectral sum includes integrals over continuum states. Each momentum takes the dimer S wave into an $l = 1$ state, and their scalar contraction selects total orbital angular momentum zero. The first-order shift vanishes, while the second-order shift is attractive. The intermediate states need not be bound P waves. The coefficient has units of energy, and its momenta contain the explicit $\hbar$ above.

This is a correction from the motion of the same retained carrier. It does not insert a direct pair potential between the endpoints. The product trial state already proves binding; the correlated wavefunction lowers its energy further. Recoil correlations are an actual dynamical effect even though their first-order energy expectation vanishes.

For subsequent orders, let $\mathcal{Q} = 1 - |0\rangle\langle 0|$ and $\mathcal{R}_0 = \mathcal{Q}(E_0 - H_0)^{-1}\mathcal{Q}$. With intermediate normalization and $|\psi\rangle = \sum_{n\geq 0}\eta^n|\psi_n\rangle$, one has

$$E_n = \langle 0|W|\psi_{n-1}\rangle, \tag{12}$$

$$|\psi_n\rangle = \mathcal{R}_0\left[W|\psi_{n-1}\rangle - \sum_{j=1}^{n-1} E_j|\psi_{n-j}\rangle\right], \qquad n \geq 1. \tag{13}$$

Odd energy coefficients vanish. This is an all-order prescription, not an evaluation of every coefficient or a claim of convergence up to $\eta = 1$. A small dimer gap can limit its useful domain.

The fixed-μ qualification matters: changing M at fixed endpoint mass m also changes μ. The dimer contribution $2\epsilon_2(\mu)$ then changes, and the total physical heavy-mass expansion need not contain only even powers of $1/M$.

4 An analytic deep-well calculation through order $\hbar^2$

For the specified Gaussian contact take $\gamma \gg 1$. Define

$$\omega^2 = \frac{2V_0}{\mu R^2}, \qquad \varepsilon = \frac{\hbar}{\mu\omega R^2} = \gamma^{-1/2} \ll 1, \qquad S_\eta = \sqrt{1+\eta} + \sqrt{1-\eta}. \tag{14}$$

Hold $0 \le \eta < 1$ fixed in this expansion. Deep binding relative to E_R can coexist with nonrelativistic control if V_0 remains small compared with the constituent rest energies and R is sufficiently large.

In $\mathbf{u}_\pm = (\mathbf{x} \pm \mathbf{y})/\sqrt{2}$, the harmonic Hamiltonian is

$$H_{\rm harm} = -2V_0 + \sum_{\sigma=\pm} \left[\frac{(1+\sigma\eta)p_\sigma^2}{2\mu} + \frac{\mu\omega^2 u_\sigma^2}{2} \right]. \tag{15}$$

Its frequencies are $\omega\sqrt{1\pm\eta}$, and hence

$$E_{3,\rm harm} = -2V_0 + \frac{3\hbar\omega}{2} S_\eta. \tag{16}$$

This resums recoil exactly within the harmonic approximation. Relative to twice the harmonic dimer energy,

$$E_{3,\rm harm} - 2\epsilon_{2,\rm harm} = -\frac{3\hbar\omega}{8}\eta^2 - \frac{15\hbar\omega}{128}\eta^4 + O(\hbar\omega\eta^6). \tag{17}$$

The first anharmonic term is $V_4 = -V_0(x^4 + y^4)/(2R^4)$, where $x^4 = (\mathbf{x}^2)^2$. The variance of each Cartesian component of $\mathbf{x}$ or $\mathbf{y}$ in the harmonic ground state is $s = \hbar S_\eta/(4\mu\omega)$. Since $\langle x^4 \rangle = 15 s^2$, first-order perturbation theory in V_4 gives

$$\boxed{E_3 = -2V_0 + \frac{3\hbar\omega}{2} S_\eta - \frac{15\hbar^2}{32\mu R^2} S_\eta^2 + O(V_0\varepsilon^3).} \tag{18}$$

Thus the classical well depth, harmonic zero-point energy and first anharmonic correction are explicitly orders $\hbar^0, \hbar^1, \hbar^2$ at fixed V_0, R, μ, η. Sextic first-order perturbation and quartic second-order perturbation both first enter at order $\hbar^3$; retaining only one would be incomplete.

The corresponding dimer expansion is

$$\epsilon_2 = -V_0 + \frac{3\hbar\omega}{2} - \frac{15\hbar^2}{16\mu R^2} + O(V_0\varepsilon^3). \tag{19}$$

In particular, the coefficient of η^2 in the combined expansion is

$$E_3 - 2\epsilon_2 = \left[-\frac{3\hbar\omega}{8} + \frac{15\hbar^2}{32\mu R^2} + O(\hbar\omega\varepsilon^2) \right] \eta^2 + O(\hbar\omega\eta^4). \tag{20}$$

The attractive curvature agrees with Eq. (11) in the controlled regime $\varepsilon \ll 1$. These formulas concern the chosen finite Gaussian regulator; they are not universal coefficients of every short-range force.

5 The whole-state current with physical momentum transfer

Take an electrically neutral λ and endpoint charges e_1, e_2 with $e_1 + e_2 \neq 0$. Equal endpoint masses and symmetric dynamics make their one-body elastic form factors equal, even when their charges differ. A

charged state may require different endpoint flavors; the binding proof did not require a nonzero electric charge.

The leading constituent current measures

$$\mathbf{r}_q - \mathbf{R}_{\rm cm} = \frac{(m+M)\mathbf{x} - m\mathbf{y}}{2m+M} = \frac{\mathbf{x}-\eta\mathbf{y}}{1+\eta}, \qquad F(\mathbf{Q}) = \left\langle e^{i\mathbf{Q}\cdot(\mathbf{r}_q - \mathbf{R}_{\rm cm})/\hbar} \right\rangle. \tag{21}$$

Here $\mathbf{Q}$ is physical momentum transfer, not wave number. With $r_E^2 = \langle |\mathbf{r}_q - \mathbf{R}_{\rm cm}|^2 \rangle$,

$$F(Q) = 1 - \frac{Q^2 r_E^2}{6\hbar^2} + O\left(\frac{Q^4 \langle r^4 \rangle}{\hbar^4}\right). \tag{22}$$

For a neutral total state the normalized electric form factor is unavailable; other currents or multipoles must be specified.

The harmonic correlated state gives

$$\langle x^2 \rangle_{\rm harm} = \frac{3\hbar}{4\mu\omega} S_\eta, \qquad \langle \mathbf{x}\cdot\mathbf{y} \rangle_{\rm harm} = \frac{3\hbar}{4\mu\omega} D_\eta, \tag{23}$$

$$D_\eta = \sqrt{1+\eta} - \sqrt{1-\eta}, \qquad r^2_{E,\rm harm} = \frac{3\hbar[(1+\eta^2)S_\eta - 2\eta D_\eta]}{4\mu\omega(1+\eta)^2}. \tag{24}$$

The positive cross-correlation is generated by carrier recoil. It reduces the mean-square radius relative to a distribution with the same marginal variance but zero cross-correlation. The total change also includes the change in that marginal variance and in the center-of-mass weights.

The Gaussian harmonic result is

$$F_{\rm harm}(Q) = \exp\left[-\frac{Q^2 r^2_{E,\rm harm}}{6\hbar^2}\right]. \tag{25}$$

For a single dressed current insertion, after matching the point-particle kinematics and assuming no independent magnetic or transition form factors, the Born rate ratio is $|F|^2$. The illustrative condition $|F|^2 \geq 1-\delta$ gives

$$r^2_{E,\rm harm} \leq -\frac{3\hbar^2}{Q^2}\log(1-\delta), \qquad 0 < \delta < 1. \tag{26}$$

Contact binding alone therefore does not remove size from the current. In particular, a mean-square radius proportional to $\hbar$ is insufficient by itself for pointlike scattering at fixed physical Q: the resolution length is $\hbar/Q$. The harmonic approximation to the current is controlled for fixed $Q\sqrt{r_E^2}/\hbar$ in the deep-well expansion, especially near the pointlike window. Its exponentially small large-Q tail is not a uniform prediction of the finite Gaussian well as $\hbar \to 0$.

5.1 Leading anharmonic radius

The first correction to the current's slope requires the changed bound state. It can be obtained without a spectral sum by adding independent quadratic sources $j_\pm u_\pm^2/2$, writing $k_\pm = \mu\omega^2 + j_\pm$, and holding the physical quartic coefficient $-V_0/(2R^4)$ fixed. Define the component variances

$$v_\pm = \frac{\hbar}{2}\sqrt{\frac{1\pm\eta}{\mu k_\pm}}, \qquad s = \frac{v_+ + v_-}{2}. \tag{27}$$

The harmonic energy depending on these sources is $(3\hbar/2)\sum_\pm \sqrt{(1\pm\eta)k_\pm/\mu}$, and $E_4 = -15V_0 s^2/R^4$. Hellmann–Feynman differentiation gives

$$\langle u_\pm^2 \rangle = 2\frac{\partial(E_{\rm harm} + E_4)}{\partial k_\pm} = 3v_\pm + \frac{15V_0 s v_\pm}{k_\pm R^4} + O(R^2\varepsilon^3). \tag{28}$$

Evaluate at $k_+ = k_- = \mu\omega^2 = 2V_0/R^2$. In the full finite well these derivatives may be taken from nonnegative added sources, avoiding an unbounded negative quadratic potential at infinity. Exchange symmetry makes $\langle \mathbf{u}_+ \cdot \mathbf{u}_- \rangle = 0$. Both mode variances, and hence both $\langle x^2 \rangle$ and $\langle \mathbf{x} \cdot \mathbf{y} \rangle$, acquire the same relative correction $5S_\eta\varepsilon/8$. Thus

$$\boxed{r_E^2 = r_{E,\mathrm{harm}}^2 \left(1 + \frac{5S_\eta}{8} \frac{\hbar}{\mu\omega R^2}\right) + O(R^2\varepsilon^3).} \tag{29}$$

The leading radius is order $\hbar$ and this correction is order $\hbar^2$ at fixed physical well parameters and mass ratio. The coefficient is positive: the finite Gaussian well is softer than its quadratic approximation and increases the size. It does not supply an additional pointlike mechanism. In the independent-dimer limit $\eta = 0$,

$$r_E^2 = \frac{3\hbar}{2\mu\omega} \left(1 + \frac{5\hbar}{4\mu\omega R^2}\right) + O(R^2\varepsilon^3), \tag{30}$$

which agrees with the single three-dimensional Gaussian well. The coefficients are at fixed $0 \le \eta < 1$; no uniform limit at $\eta = 1$ is asserted. Equation (29) improves the small-Q expansion (22), under the same charge assumptions. Substituting this radius into a Gaussian exponential would not derive the anharmonic finite-Q form factor: its higher moments change as well.

6 Contact iterations are not the same as the semiclassical series

Choose momentum normalization $\langle \mathbf{r}|\mathbf{p}\rangle = e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}$, with completeness measure $d^3p/(2\pi\hbar)^3$. The dimer potential kernel is

$$V(\mathbf{p}', \mathbf{p}) = -\frac{3G}{2} \exp\left[-\frac{R^2|\mathbf{p}' - \mathbf{p}|^2}{4\hbar^2}\right]. \tag{31}$$

Its scattering operator obeys

$$T(\mathbf{p}', \mathbf{p}; E) = V(\mathbf{p}', \mathbf{p}) + \int \frac{d^3k}{(2\pi\hbar)^3} \frac{V(\mathbf{p}', \mathbf{k})\, T(\mathbf{k}, \mathbf{p}; E)}{E - k^2/(2\mu) + i0}. \tag{32}$$

Every factor has physical units: V and T here have units of energy times volume. Repeated propagation in this equation is organized by the contact strength and the energy denominator, not by assigning an extra positive power of $\hbar$ to every iteration. At fixed physical G, R, μ, $\gamma \propto \hbar^{-2}$; taking $\hbar$ small makes this interaction strong in dimensionless Schrödinger units.

A bound-state pole is a nonperturbative feature of the full resolvent. Quantum-field-theory loop counting instead follows from a specified action in $\exp(iS/\hbar)$ and a stated field/coupling normalization. The present fixed-number Hamiltonian has no vacuum pair-creation loops. A relativistic completion would have to specify them before their $\hbar$ orders could be calculated. We have not computed such a completion's radiative corrections.

7 Result and next calculation

The retained adjoint contact model binds both endpoints without a direct endpoint force. Keeping $\hbar$ explicit reveals three separate controls, attractive second-order recoil, and the deep-well energy and whole-state radius through $\hbar^2$. The leading anharmonic radius correction is positive. This crosses the current calculation's milestone without establishing the desired experimental pointlikeness.

The next quantitative target is the first anharmonic correction to the finite-momentum current for fixed $Q\sqrt{r_{E,\mathrm{harm}}^2}/\hbar = O(1)$, including its non-Gaussian shape. This will test a declared resolution

interval rather than only its zero-momentum slope. A different near-threshold target would require resummed two- and three-body propagation. Neither needs an immediate three-generation model. A string or Seiberg-dual completion remains a possible bridge to the broader correspondence, with an explicit operator dictionary and physical currents.

The previous scalar-completion study produces spin-dependent contacts rather than the spin-independent interaction assumed here. For physical spins its projectors are $P_0 = 1/4 - \mathbf{s}_i \cdot \mathbf{s}_j/\hbar^2$ and $P_1 = 1 - P_0$. Its binding proof cannot be imported unchanged. This limitation does not alter the result for the Hamiltonian in Eq. (4).

8 Endpoint statistics and a charged supercharge: construction hypotheses

The string interpretation leaves an additional choice open. In an RNS construction the physical NS/R sectors determine the whole string's spacetime statistics; Chan–Paton labels alone do not specify independently propagating spin-1/2 endpoint particles. Physical endpoints require boundary fields and an action. The contact model above chooses two fermionic endpoints, while the user now also proposes two scalar endpoints for the fermionic string. Either pair has even fermion parity and can give an odd state with an odd retained carrier. Mixed scalar/fermion endpoints plus an odd carrier instead have even total parity. The ordinary SQCD mesino is odd with mixed endpoints and no additional odd carrier. These are distinct candidates to match, not interchangeable descriptions already derived from a common theory.

Three charged Dirac leptons have the spin degrees of freedom of six complex spin-zero fields. The six charged pseudoscalar mesons formed from two up-type and three down-type hadronizing flavors provide this count. Direct antecedents exist: Rivero's 2005 composite-boson proposal and 2009 *sBootstrap* discussion explicitly pursue the three-generation correspondence and the pointlike-lepton difficulty [7, 8]. The present discussion continues those proposals; neither the count nor this research direction is claimed as new here. Candidate pairs are $\mu \leftrightarrow (\pi, K)$, $\tau \leftrightarrow (D, D_s)$ and $e \leftrightarrow (B, B_c)$, with charge conjugates understood. A dynamical supersymmetry representation, mass splittings and the top's absence from the ordinary meson list remain to be explained.

8.1 The pion/kaon pair and a six-meson mixing space

The four charged states $\pi^\pm, K^\pm$ do supply the scalar field content for one charged Dirac multiplet. A formal pair of left-chiral multiplets is

$$\Phi_- = (\pi^-, \mu_L^-), \qquad \Phi_+ = (K^+, (\mu_R^-)^c), \tag{33}$$

with auxiliary fields suppressed and conjugates understood. The two Weyl fields form the Dirac muon. This is a viable count and electric-charge assignment; exact unbroken supersymmetry would additionally constrain masses and interactions. The $B - L$ constraint below is an additional premise.

Allow the partner basis to involve all six charged mesons. For canonically normalized effective scalar fields define

$$P^- = (\pi^-, K^-, D^-, D_s^-, B^-, B_c^-)^T, \qquad \begin{pmatrix} A^- \\ C^- \end{pmatrix} = UP^-, \quad U \in U(6), \qquad B_i^+ = (C_i^-)^*. \tag{34}$$

The two three-component blocks furnish candidate scalar coordinates A_i^- and B_i^+ for three pairs of left-chiral multiplets. This parameterizes the pairing rather than fixing the flavor basis as the partner basis. Dynamics must determine the kinetic normalization, allowed flavor breaking, mass matrices and

U; no angles or masses are fitted here. Because ordinary mesons all have $X = B - L = 0$, $UX_PU^\dagger = 0$ for $X_P = 0_6$. Mixing may change the partner assignments but cannot alone repair a missing nonzero X charge. These are dimensionless representation equations, with no convention $\hbar = 1$.

8.2 A common $B - L$ shift

For conventional dimensionless number charges, let $X = B - L$. A candidate supercharge $\mathcal{Q}_+$ can have the same charge shift in both branches:

Transition	Initial X	Final X	Shift
meson $\to$ antilepton	0	$+1$	$+1$
antidiquark $\to$ quark	$-2/3$	$+1/3$	$+1$
meson $\to$ lepton	0	-1	-1
diquark $\to$ antiquark	$+2/3$	$-1/3$	-1

The first two transitions use $\mathcal{Q}_+$ and the last two its conjugate $\mathcal{Q}_-$. The corresponding color representations agree. With suitable flavors the electric charges can agree as well. Algebraically,

$$[X, \mathcal{Q}_\pm] = \pm\mathcal{Q}_\pm, \qquad [\hbar X, \mathcal{Q}_\pm] = \pm\hbar\mathcal{Q}_\pm. \tag{35}$$

The second expression makes explicit the convention for a dimensionful generator; the first does not set $\hbar$ to one. If these are spacetime supercharges, X acts as an R symmetry. Their conjugate charges allow a neutral translation generator in their anticommutator. This is an admissible charge ledger, not yet a representation or interaction law.

The two positive shifts require respectively $(\Delta B, \Delta L) = (0, -1)$ and $(1, 0)$. One fixed charge eigenoperator cannot implement both while carrying definite charges under independently conserved B and L. In a retained-carrier realization, an exact nonzero continuous X charge also requires distinguishable carrier/anticarrier sectors; a self-conjugate Majorana carrier cannot carry it.

There is a further, useful constraint on a *linear, exact* $N = 1$ interpretation. Choose the convention that a left-chiral scalar of charge x has Weyl partner charge $x-1$. A charged Dirac lepton uses left-handed Weyl fields with $X = -1, +1$, so the corresponding scalars have $X = 0, +2$. Equivalently the two scalar particles with the lepton's electric charge have $X = 0, -2$. Two ordinary mesons, both $X = 0$, do not fill this multiplet with conserved physical $B - L$. Possible construction choices include an additional scalar with two units of lepton number, breaking or reassigning the R charge, or a nonlinear/emergent supersymmetry without these simple observed-state multiplets. The constraint has these explicit premises; it is not an exclusion of the broader string proposal. Making physical $B - L$ an ordinary symmetry commuting with the supercharge would instead demand equal scalar and fermion $B - L$, also differing from the ordinary meson/lepton assignment. The conserved symmetry must be specified explicitly in either construction.

8.3 A historical mass-protection mechanism and endpoint charges

Greenberg, Mohapatra and Yasue's supplied 1983 paper [9] uses supersymmetric preon dynamics and chiral symmetry to protect light composite quarks and leptons. Its quasi-Nambu–Goldstone fermions are partners of Goldstone bosons; they are not necessarily Goldstinos of broken supersymmetry. Their six flavor–color labels form a $4 + 2$ partition in $SU(6)$, distinct from the six charged mesons here. With their assumed condensates and unbroken group, the anomaly condition is

$$2(l_1 - l_2 + l_3) = 6. \tag{36}$$

Three generations follow on choosing $l_2 = l_3 = 0$ and $l_1 = 3$; the equation alone does not select that solution. Their illustrative generation mass matrix is not a six-meson mixing matrix. The transferable idea is symmetry protection of a light composite fermion. It supplies no whole-state form factor or automatic pointlike behavior.

The paper's residual R-type charge, denoted R_A here to distinguish it from physical $B - L$, assigns charges $0, +1, -1$ to scalar matter, left-handed matter fermions and the adjoint fermion at $N = N_f = 6$. Applying that convention to supersymmetric gauge-theory operator analogues gives

$$\chi_{M,\alpha} = \widetilde{A}\psi_\alpha + \widetilde{\psi}_\alpha A, \qquad R_A(\chi_M) = +1, \tag{37}$$

$$H_{F,\alpha} = (\widetilde{\psi}^\beta T^a \psi_\beta)\lambda^a_\alpha, \qquad R_A(H_F) = +1, \tag{38}$$

$$H_{S,\alpha} = (\widetilde{A} T^a A)\lambda^a_\alpha, \qquad R_A(H_S) = -1. \tag{39}$$

Tilded fields are independent chiral matter in the conjugate gauge representation, not complex conjugates. All three operators have odd fermion parity. The source's $SU(6)$ hypercolor is distinct from the active contact model's $SU(3)$ color; this table does not identify the two theories. The matching charges of H_F and χ_M permit but do not calculate an overlap. In an R_A-invariant vacuum, $\langle H_F(x) H_S^\dagger(0)\rangle$ vanishes by charge conservation, while the mass-type product $H_F^\alpha H_{S,\alpha}$ is R_A-neutral. A physical mass mixing also requires conjugate electric and other conserved charges; this has not been supplied by the R-charge count alone. Opposite R charges therefore do not forbid every coupling. The source later assumes breaking of the remaining U(1) factors, which changes the selection rules. Boundary dynamics, mass protection and the current still require a common construction; these charge sums are not a binding calculation.

8.4 The pion–muon–Goldstino conjecture

Volkov and Akulov proposed a neutrino as the Goldstone fermion of nonlinear supersymmetry [10]. Their supplied two-page paper does not identify a pion–muon partner pair. We retain the additional conjecture

$$\pi^- \longrightarrow \mu^- + \bar{G}, \qquad \bar{G} \text{ tentatively identified with } \bar{\nu}_\mu, \tag{40}$$

as a broken-supersymmetry scalar-partner transition. The assignments $X(\mu^-) = -1$ and $X(\bar{G}) = +1$ conserve $B - L$; the emitted state is electrically neutral. Nonlinear supersymmetry does not itself require a light scalar partner, so the proposed pion identification is additional physical content. Its next substantive test would specify the Goldstino coupling and compare pion leptonic amplitudes, their helicity dependence and the muon/electron channel ratio. No such amplitude or decay rate is calculated in this revision.

9 Partial compositeness and the sBootstrap comparison

The supplied v1 sources of Agashe, Ricci and Sundrum [3] and Agashe, Giudice, Rattazzi and Sundrum [4] provide two related approaches to hierarchical flavor. Both couple elementary matter to operators of a new strong sector. Their operator dimensions generate hierarchical mixing strengths from structureless ultraviolet values. In the irrelevant-coupling, approximately fixed-point regime,

$$\widehat{\lambda}_i(E) = \widehat{\lambda}_i(\Lambda_{\mathrm{UV}}) \left(\frac{E}{\Lambda_{\mathrm{UV}}}\right)^{\gamma_i}. \tag{41}$$

Here both scales are energies and the couplings are dimensionless; this relation introduces no convention setting physical $\hbar$ to one. Relevant mixings, including those needed for the large top Yukawa, require the nonlinear RG treatment of the sources. The nonsupersymmetric fermionic-operator exponent is $\gamma = \Delta_f - 5/2$, whereas the supersymmetric chiral-superfield exponent is $\gamma = \Delta_s - 2$. The chiral-multiplet relation $\Delta_f = \Delta_s + 1/2$ makes these the same scaling criterion.

Different protection mechanisms. The 2025 construction generates hierarchies in a flavor-anarchic UV stage and then assumes a transition to strong dynamics with emergent flavor and CP symmetries. Right-handed mixings flow towards universal values, while left-handed mixings retain the flavor hierarchies. Its MFV example admits compositeness around 10 TeV, with a less symmetric variant around 5 TeV. A significant consequence is that right-handed fermions of *all* generations can have sizable compositeness: light mass does not require both chiralities to be nearly elementary. The Higgs is protected as a pseudo-Nambu–Goldstone boson.

The 2026 Super-Composite Higgs instead retains a flavor-anarchic strong sector and combines it with supersymmetry. Composite Higgs superfields emerge at a scale m_*, residual strong dynamics breaks SUSY at a lower scale m_S, and superpartner masses occur at the soft scale. These source symbols denote energy scales. Supersymmetry permits a higher flavor/compositeness scale while protecting Higgs lightness; holomorphy protects a vanishing supersymmetric μ term until SUSY breaking. Gauge-mediation-like soft terms and restrictions on supersymmetric operators help control flavor effects. Compositeness also supplies the proposed μ–B_μ mechanism. Thus this is not simply the 2025 emergent-flavor construction with SUSY added. The analyzed scales extend to multi-PeV compositeness, and the authors discuss electron-EDM, gravitino and collider signatures with model and cosmological assumptions. Residual tuning remains.

Comparison with the original sBootstrap. Rivero's proposal [7, 8] identifies ordinary QCD mesons and diquark/antidiquark channels as scalar-partner candidates. This describes the earlier hadronic identification used in this notebook. The subsequently supplied 2024 paper explicitly calls for different high-energy bound states; Section 10 corrects and refines that comparison. Its six charged meson flavor states and conjugates arise from the five hadronizing flavors u, c, d, s, b, with top excluded. The two new papers instead posit a new strong sector and take three SM generations and suitable composite-operator families as inputs. Their six up/down operator channels, where present, are not a derivation of those six QCD mesons. Substantial top compositeness under a new force also differs from top's inability to form ordinary QCD mesons.

An ordinary spin-zero meson cannot be the fermionic operator in a linear elementary-lepton mixing. A suitable odd operator, its quantum numbers and its overlap must be supplied. The retained-adjoint operator studied in this notebook is a candidate hypothesis, not an operator identified or dynamically derived by either paper. Moreover, in the 2026 conventional supersymmetric portal, elementary and composite fields mix as whole superfields. In an unbroken, canonically normalized linear realization, the scalar and fermion components share that mixing rotation. It does not independently choose a wholly QCD-composite scalar and an elementary fermion. This statement concerns that realization, not every possible emergent or nonlinear supersymmetry.

There is also a charge distinction: ordinary mesons have $B = L = 0$, whereas conventional sleptons share their lepton partners' lepton number. The 2026 assumptions preserving B and L do not implement the sBootstrap identification. The earlier conditional physical $B - L/\mathrm{R}$ analysis remains relevant. Finally, an SQCD meson chiral superfield has a scalar–scalar lowest component and a scalar–fermion fermionic component. Identifying it with a QCD pseudoscalar built from fermionic quarks requires physical operator matching, not just the shared word "meson."

What transfers and what remains open. The useful ingredients are RG-generated mixing hierarchies, emergent flavor protection, and separation of composite-state formation from SUSY breaking. The 2026 hidden-SQCD/ISS toy illustrates staged dynamics but explicitly has an unrealistic gauge-mediated spectrum with suppressed gaugino masses. Neither paper supplies a fully derived realistic microscopic completion or a QCD sBootstrap vacuum. Its Goldstino belongs to a SUSY-breaking sector and becomes the gravitino's longitudinal modes; it is not identified with a neutrino.

Neither work calculates the whole-state current of the proposed fermionic color string or proves that it is pointlike while its bosonic partner is extended. Small mixing or a high new-physics scale can suppress observable compositeness, but those mechanisms require an explicit comparison with this notebook's target. The next construction task would specify one candidate lepton operator and scalar partner, their conserved charges and SUSY realization, then their current overlaps. No new finite-Q result or flavor fit is claimed in this source-comparison step.

10 The 2024 sBootstrap and the SO(32) completion

The subsequently supplied paper by Rivero [1] makes an important refinement to the preceding comparison. Sections 5 and 8 explicitly distinguish its proposed scalar composites from ordinary QCD-scale mesons and diquarks, and envisage a different high-energy binding mechanism. Thus the distinctive hypothesis is that a subset of observed SM fermions supplies the constituents of scalar partners; it is not necessarily an identification of those partners with known QCD hadrons. This narrows the contrast with new strong dynamics in Refs. [3, 4], although the constituent and operator dictionaries remain different.

What the five-flavor count supplies. Let $V = (5, 3)$ under $SU(5)_f \times SU(3)_c$. Its complex dimension is 15, so its underlying real space has dimension 30. The natural orthogonal extension is therefore $SO(30)$. Its adjoint already contains the channels used in the scalar count:

$$435 = 224 + 105 + \overline{105} + 1, \tag{42}$$

$$224 = (24, 1) + (1, 8) + (24, 8), \tag{43}$$

$$105 = (15, \bar{3}) + (10, 6). \tag{44}$$

Indeed Sections 3.2 and 7.1 of Ref. [1] explicitly recognize $SO(30)$ as sufficient for the relevant decompositions.

The minimal extension to 32 real labels is one additional complex color–flavor singlet:

$$16 = (5, 3) + (1, 1), \qquad 32_{\mathbb{C}} = (5, 3) + (\bar{5}, \bar{3}) + 1 + 1. \tag{45}$$

These are internal representation dimensions, not counts of spinor components. A sterile-neutrino-like endpoint and its conjugate are a possible physical interpretation, not a result of the count. Here the fifteen labels are five quark flavors times three colors, rather than the fifteen Weyl fermions of one SM family. The familiar right-handed neutrino completion of a different family embedding therefore does not automatically apply. Adding the omitted colored top instead gives 18 complex labels and points to $SO(36)$.

The extra adjoint content. Taking the exterior square of $\mathbb{R}^{30} \oplus \mathbb{R}^2$ gives the branching under $SO(30) \times SO(2)$,

$$496 = (435, 1) + (30, 2) + (1, 1). \tag{46}$$

The difference is 61 adjoint components, not just two extra scalar states. In color–flavor language the added content is $2(5, 3) + 2(\bar{5}, \bar{3}) + (1, 1)$, precisely the mixed channels and additional singlet visible in Eq. (1) of Ref. [1]. A candidate neutral endpoint must account for these mixed colored channels. Its electric charge also needs an explicit embedding. All these identities are dimensionless and make no convention setting physical $\hbar$ to one.

An enlargement is not yet a selection rule. The paper suggests replacing color $SU(3)$ by $U(3)$. However, the unitary action $A \otimes B$ of $SU(5) \times U(3)$ on $\mathbb{C}^{15}$ already acts orthogonally on $\mathbb{R}^{30}$, with real determinant $|\det_{\mathbb{C}}(A \otimes B)|^2 = 1$. Its image, up to a finite kernel, lies in $U(15) \subset SO(30)$. The extra central phase alone consequently does not force 32 real directions. Nor does adding singlets by itself demonstrate interactions invariant under the full larger algebra.

One possible additional selector is the standard ten-dimensional $N = 1$ supergravity/super-Yang–Mills Green–Schwarz anomaly-cancellation condition. Its gravitational anomaly requires 496 gauge generators; within the orthogonal family,

$$\frac{n(n-1)}{2} = 496 \quad \Longrightarrow \quad n = 32. \tag{47}$$

Gauge and mixed anomaly factorization are also required; the type-I string implements a related selection through tadpole cancellation. This is a conditional higher-dimensional input, not a consequence of the four-dimensional scalar count. The presence of 496 scalar components alone does not supply the necessary ten-dimensional gauge sector, and $E_8 \times E_8$ is another familiar string possibility.

The missing ingredient is therefore a physical reason for exactly one complex neutral sector and for the full symmetry, or a genuine string-consistency condition selecting it. A neutral constituent is a useful guess; no particular particle assignment, electroweak completion or binding dynamics is derived here.

11 Scalars as internal components of the ten-dimensional gauge field

Section 10 left two questions open: what removes the adjoint components outside the selected channels, and why a scalar multiplet with 496 entries should obey a condition on the ten-dimensional gauge sector. This section tests one hypothesis that addresses both: the scalars of Ref. [1] are the components of the ten-dimensional $SO(32)$ gauge field along compact directions. All statements are conditional on that identification. Except for Eq. (48), they are dimensionless representation statements. Details, scripts and outputs are in blackboard 012 of the research notebook.

Labels. Fifteen of the sixteen Cartan directions carry a pair (flavor, color), with flavor in $\{u, c, d, s, b\}$; the sixteenth, n, is colorless. The roots $\pm e_a \pm e_b$ then split into meson-like roots $e_{(i,c)} - e_{(j,d)}$, diquark-like roots $e_{(i,c)} + e_{(j,d)}$, which carry $(15, \bar{3}) + (10, 6)$, and one-label roots $e_{(i,c)} \pm e_n$. Electric charge is the Cartan generator $Q = \sum_a q_a H_a$ with the quark charges on the colored labels; it reproduces the charges of Ref. [1] on the selected channels.

Nine dimensions. The nine-dimensional $N = 1$ vector multiplet contains one real scalar. For a string theory on a circle it is the gauge-field component A_9, in the 496. Restricted to $(24, 1) + (15, \bar{3}) + (\overline{15}, 3)$ it has $24 + 2 \times 45 = 114$ real components: twelve for charged and twelve for neutral slepton-like states, and, per color, six complex states of charge 1/3, six of charge 2/3 and three of charge 4/3. This is the count of two scalars per Dirac fermion of Ref. [1] with no additional copies. Each further compact dimension adds one real scalar per generator. Reference [8] had singled out nine dimensions as the non-chiral stage. Under the hypothesis the scalar multiplet is the gauge sector, so the ten-dimensional anomaly condition of Section 10 becomes the selector of $SO(32)$ over $SO(30)$.

Label symmetry. In the fermionic formulation of the heterotic string the 32 labels are carried by left-moving worldsheet fermions λ^I, and the adjoint states are the bilinears $\lambda^I_{-1/2}\lambda^J_{-1/2}|0\rangle$, antisymmetric by worldsheet statistics. Space-time spin comes from the right-movers. The channel $(15, \bar{3}) = S^2(5) \otimes \Lambda^2(3)$ is therefore a bilinear of two label-carrying fermions with no constraint from space-time statistics. The left-moving internal central charge 16 corresponds to sixteen complex labels, $5 \times 3 + 1$.

Mass formula. With a Wilson line a on a circle of radius R, a state on root p has, at the lowest Kaluza–Klein level,

$$M = \frac{\hbar\,|p\cdot a|}{R\,c_{\text{light}}}, \qquad M^2 \propto (a_a \mp a_b)^2 \tag{48}$$

for meson-like and diquark-like roots respectively. The self-energy formula $E \propto (q_a + q_b)^2$ of Ref. [1] has this form if E is read as M^2, with the condition $z_1 + z_2 + z_3 = 0$ a Wilson line in the Cartan subalgebra of the $SU(3)$ acting on (d, s, b). The second condition of that formula is an additional input. A label reflection $e_a \to -e_a$ is a Weyl reflection and exchanges meson-like and diquark-like sectors. In nine dimensions a scalar on a root with $p \cdot a \neq 0$ is absorbed by the massive vector.

Orbifold projections. For $T^6/\mathbb{Z}_N$ orbifolds preserving four-dimensional $N = 1$, with heterotic consistency conditions, no Wilson lines and color-blind shifts $V = (w_u^3, w_c^3, w_d^3, w_s^3, w_b^3, w_n)$, a root gives a gauge boson if $p \cdot V \in \mathbb{Z}$, one chiral multiplet from A_{z_k} if $p \cdot V = v_k \bmod 1$, and is projected out otherwise. An exact enumeration gives the following.

- $\mathbb{Z}_3$ reproduces the series $U(3k) \times SO(32 - 6k)$. Its member with gauge group $U(15) \times SO(2)$ has untwisted matter $3\,[\overline{10}_{5} + 15_{+1} + 15_{-1}]$ and twisted matter $27\,[15_{-1/2} + 3 \cdot 1]$: three copies of the diquark-like multiplet, one per complex plane.
- Single copies occur. With the split $(u, c \,|\, d, s, b)$, shifts in $\mathbb{Z}_8$-I and $\mathbb{Z}_{12}$-I give $U(6) \times U(9) \times U(1)$ with every diquark-like label as one chiral multiplet and the charged meson-like labels projected out.
- The $\mathbb{Z}_7$ shift $V = (1^6, 2^9, 0)/7$ gives $U(6) \times U(9) \times SO(2)$ with every charged label exactly once (Table 1); the twelve neutral meson-like labels are gauge bosons of the flavor subgroup.
- No admissible shift isolates the selected channels: $(10, 6)$ accompanies $(15, \bar{3})$ and $(24, 8)$ accompanies $(24, 1)$. Among the 96 shifts that make all fifteen diquark-like labels chiral with one orientation, none keeps a meson-like label as matter.

Cubic non-abelian anomalies cancel between untwisted and twisted sectors in the five non-trivial $\mathbb{Z}_3$ models and in the $\mathbb{Z}_7$ model, which fixes the orientation convention. Twisted sectors of the non-prime orbifolds were not computed.

Table 1: Massless charged spectrum of the $\mathbb{Z}_7$ orbifold with $V = (1^6, 2^9, 0)/7$. Untwisted states are internal components of the ten-dimensional gauge field.

Sector	Copies	$U(6) \times U(9)$, $SO(2)$ charge	Labels
untwisted, plane 1	1	$(\bar{6}, 9)$	charged mesons $d\bar{u}$
untwisted, plane 1	$1 + 1$	$(6, 1)_{\pm 1}$	$u\,n$, $u\,\bar{n}$
untwisted, plane 2	1	$(15, 1)$	uu, charge $4/3$
untwisted, plane 2	$1 + 1$	$(1, 9)_{\pm 1}$	$d\,n$, $d\,\bar{n}$
untwisted, plane 3	1	$(1, 36)$	dd, charge $-2/3$
untwisted, plane 3	1	$(\bar{6}, \bar{9})$	$\bar{u}\bar{d}$, charge $-1/3$
twisted θ	21	singlets	
twisted θ^2	7, 7, 21	$(6, 1)$, $(1, \bar{9})$, singlets	
twisted θ^4	7, 7	$(6, 1)$, singlets	

The sixteenth label. The 61 generators that distinguish $SO(32)$ from $SO(30)$ are the states with exactly one colored label and one singlet. With $\mathfrak{so}(32) = \mathfrak{k} \oplus \mathfrak{p}$, $\mathfrak{k} = \mathfrak{so}(30) \oplus \mathfrak{so}(2)$ and $\mathfrak{p} = (30, 2)$, one has $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$: two one-label states compose into a meson-like or diquark-like label. The charge formula of Ref. [1] gives the new label charge zero. If instead Q is required to be traceless on the sixteen complex labels, the colored labels sum to $3(2/3 + 2/3 - 1/3 - 1/3 - 1/3) = 1$ and the new label has charge -1. The one-label states are then leptoquark-like. Three of them, $\bar{d}\ell^+$, $\bar{s}\ell^+$, $\bar{b}\ell^+$, have color $\bar{3}$, charge $4/3$ and $B - L = 2/3$, the quantum numbers of the three uu-type states, and would complete that sector to six. A colored top would instead require eighteen labels. The tracelessness requirement is an assumption, and the same label adds two states of charge $1/3$, three of charge $2/3$ and two of charge $5/3$ in color $\bar{3}$.

What is missing. The gauge groups found are $U(15)$ or $U(6) \times U(9)$; the product structure $15 = 5 \times 3$ is a choice of basis, and no mechanism here leaves $SU(3)$ color gauged with flavor global. Ordinary supersymmetry pairs each scalar with a fermion of the same labels, whereas the sBootstrap pairs a two-label scalar with a one-label fermion. Abelian anomalies, Wilson lines and couplings were not analyzed.

12 Color from an 't Hooft twist

Section 11 ended with gauge groups $U(15)$ or $U(6) \times U(9)$, in which $15 = 5 \times 3$ is a choice of basis. This section examines one mechanism that makes the product structure physical: a flux background of the gauge field on a two-torus, the setting of Ref. [2] at rational noncommutativity. Statements about twisted bundles and index numbers are standard and are rederived in blackboard 013; the conclusions for the sBootstrap are conditional.

Twist. A $U(15)$ bundle of first Chern number $m = 3m'$, with m' prime to 5, has transition functions $U \otimes 1_3$ and $V^{m'} \otimes 1_3$ on adjoint fields, with U, V the 5×5 clock and shift matrices. Their commutant is $1_5 \otimes U(3)$: color is the unbroken group and the five-fold index is a wrapping number.

Mesons on a noncommutative torus. Expanding an adjoint field in $U^a V^b \otimes \phi_{ab}(x)$ gives the 3×3 fields ϕ_{ab} fractional momenta $b/5$ and $-m'a/5$ in units of $2\pi\hbar/L_i$. The single-valued generators $Z_1 = V e^{2\pi i x_1/5L_1}$ and $Z_2 = U^{-s} e^{2\pi i x_2/5L_2}$, with $sm' = 1 \bmod 5$, satisfy $Z_1 Z_2 = e^{2\pi i s/5} Z_2 Z_1$. The meson-like fields, read in Ref. [1] as $(24 + 1) \otimes (8 + 1)$, are therefore $U(3)$ fields on the noncommutative torus with $\theta = s/5$: the number of colors is the rank and the number of flavors is the denominator of θ. Flavor symmetry is reduced to $\mathbb{Z}_5 \times \mathbb{Z}_5$, and the lowest state of class (a, b) has

$$M = \frac{2\pi\hbar}{c_{\text{light}}} \sqrt{\Big(\frac{\tilde{b}}{5L_1}\Big)^2 + \Big(\frac{\tilde{a}}{5L_2}\Big)^2}, \tag{49}$$

with $\tilde{a}, \tilde{b} \in \{-2, \ldots, 2\}$ the representatives of $-m'a$ and b. On a square torus the 24 non-singlet classes split as $4, 4, 4, 8, 4$ at $M^2 \propto 1, 2, 4, 5, 8$; only the $U(3)$ gauge multiplet keeps a zero mode.

Diquark-like zero modes. Since $\Lambda^2(5 \otimes 3) = S^2(5) \otimes \Lambda^2(3) + \Lambda^2(5) \otimes S^2(3)$, the numbers of color-antitriplet and color-sextet zero modes on the torus are $c_1(S^2 E_5) = 6m'$ and $c_1(\Lambda^2 E_5) = 4m'$, in place of the flavor multiplicities 15 and 10. For wrapping number n they are $(n + 1)m'$ and $(n - 1)m'$.

Charge. For any background of the form $f \otimes 1_3$ the unbroken group is the unitary group of $S' \otimes M_3$, with S' the commutant of the flavor parts. A single color group needs $S' = \mathbb{C}$, which leaves no flavor-dependent charge. A charge with eigenvalues $2/3$ and $-1/3$ needs $S' \supseteq \mathbb{C} \oplus \mathbb{C}$ and gives $U(3)_u \times U(3)_d$.

The corresponding construction has two stacks with wrapping numbers 2 and 3, fluxes m_u and m_d, the label count $3\cdot 2+3\cdot 3+1=16$, and $Q=\frac{2}{3}Q_u-\frac{1}{3}Q_d+q_nQ_n$. Its zero modes on the flux torus are $3|m_u|$ antitriplets and $|m_u|$ sextets in the uu sector, $4|m_d|$ and $2|m_d|$ in the dd sector, $|3m_u+2m_d|$ ud diquarks and $|3m_u-2m_d|$ charged mesons. For $m_u=1$ the uu sector has the three antitriplets of Ref. [1]; the counts six and six of the other sectors would need $m_d=3/2$. A scan of factorized fluxes on T^6 with flux numbers between -4 and 4 finds no assignment with the ratio $1:2:2$ and none that cancels the tadpoles with these three stacks.

Assessment. The flux mechanism singles out color by turning flavor into wrapping. The multiplicities that Ref. [1] counts as dimensions of flavor representations become index numbers and momentum classes with different values, and a flavor-dependent charge requires two color groups. The counts of Ref. [1] are recovered when the flux is absent, where color is not singled out. A background outside the flavor directions, or a mechanism other than symmetry breaking, would be needed to obtain both.

Reproducibility and review scope

The radius derivation is analytic; no numerical physics script was required. A research agent derived the leading anharmonic radius and the parent independently checked only that new mathematics. The parent analyzed the conceptual $B-L$ ledger and linear-multiplet constraint. Earlier source reading and document builds were delegated to Sol. The subsequent representation-count clarification and 1983 source comparison add no finite-Q calculation. Version 0.6 added the parent's reading of supplied 2025/2026 LaTeX sources and the scoped comparison in Section 9. Version 0.7 corrects the source interpretation using the supplied 2024 paper and adds the dimensionless representation analysis in Section 10. Sol extracted that PDF and validates the document; the parent checked consequential passages on original page images. Version 0.8 adds Section 11. It was written on 10 November 2026 by Claude (model Fable 5.2) in a separate session, after reading Refs. [1, 7, 8], a 2007 preprint of the same program and Section 10. Its enumeration of orbifold shifts and its twisted spectra use exact-arithmetic scripts in `scripts/so32_gauge_scalars/`, written and run by the same model, with outputs in `calculations/so32_gauge_scalars/`. The independent consistency criteria are the reproduction of the $\mathbb{Z}_3$ series of gauge groups and the cancellation of cubic anomalies between sectors computed by different rules. No second model reviewed that section, and it was built with `pdflatex` because `latexmk` was unavailable. Version 0.9 adds Section **??**, written the same day by the same model after the user pointed to Ref. [2]; only the abstract of that reference was retrieved, and the standard results it relies on are quoted from memory and rederived in blackboard 013. Its sanity checks, tables and bounded flux scan come from `scripts/so32_gauge_scalars/flux_twist.py`. Earlier recoil, energy and current calculations are unchanged. These checks do not establish experimental correctness, novelty or a completed supersymmetric model.

The living source and PDF are in

`discussions/2026-10-11-quark-antiquark-open-string/manuscript/`

The research notebook records the model assumptions, source exposure, prior results and next task. The historical motivations include Schwarz's dual quark–gluon model [5], with Seiberg duality [6] retained as a possible later connection. Neither reference proves the contact interaction or its intended string interpretation.

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